Fast Anchor Graph Regularized Relaxation Linear Regression for Classification of Hyperspectral Images

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***Abstract*—Hyperspectral image (HSI) classification remains challenging due to the inherently high dimensionality and complex intrinsic structure of the data. These characteristics pose significant difficulties for traditional linear regression (LR) methods, particularly when it comes to capturing nonlinear relationships and preserving local structural information. To address these issues, this article proposes a novel classification framework called fast anchor graph regularized relaxation LR (FAGRLR). The proposed method integrates a fast anchor graph (AG) model into LR to encode both spatial and structural neighborhood information inherent in HSIs. By utilizing AGs, the method efficiently preserves critical local relationships within the data, enhancing the discriminative ability of the model. To further avoid overfitting, a label relaxation mechanism is introduced to increase the label matrix's flexibility in discriminative transformation learning. Additionally, the $L_{2,1}$-norm constraint is incorporated to promote robustness and sparsity in the learned transformation matrix, thereby enhancing the generalization performance of the proposed method. We validate the effectiveness of FAGRLR through extensive experiments conducted on four benchmark hyperspectral datasets. The experimental results demonstrate that the proposed method achieves superior classification performance compared to several state-of-the-art methods.**

***Index Terms*—Classification, fast anchor graph (AG), hyperspectral image (HSI), relaxation linear regression (RLR).**

I. Introduction

SINCE hyperspectral image (HSI) contains fine spatial resolution and rich spectral content, it offers substantial potential for diverse applications [1], including precision agriculture [2], land-cover classification [3], [4], [5], environmental monitoring [6], [7], [8], and mineral exploration [9], [10], [11]. HSI classification has become a pivotal area of research in remote sensing, drawing increasing interest from the machine learning (ML), image processing, and computer vision communities. In the past decades, a large amount of HSI classification methods have been developed based on classical shallow ML [12], [13], [14], and deep learning [15], [16].

Benefiting from the advantage in deep feature extraction, deep learning-based methods, including convolutional neural networks (CNNs) [17], [18], generative adversarial networks (GANs) [19], [20], and graph convolutional networks (GCNs) [21], [22], have been extensively explored in HSI classification. To further explore spatial–spectral information of HSI, a graph attention network was introduced into deep network models, resulting in methods such as weighted feature fusion of CNN and graph attention network (WFCG) [23] and graph-polarized fusion network (GPF-Net) [24]. Since Transformer architecture [25] can capture long-range spectral–spatial dependencies for HSI classification, a series of Transformer-based HSI classification methods, including DSformer [26], GraphGST [27], SSFEN [28], and DiffFormer [29], have been developed. With Mamba architecture achieved excellent performance in classification, SS-Mamba [30], IGroupSS-Mamba [31], 3DSS-Mamba [32], and MambaHSI+ [33] are proposed by integrating different spatial–spectral information extraction modules into the Mamba network. To improve the ability of model for transferring knowledge across tasks and scenes, the foundation models, including HyperSIGMA [34] and HyperFree [35], have emerged as a new research direction for HSI processing. Additionally, some lightweight models based on MLP were developed to improve the efficiency of HSI classification [36], [37]. However, the high computational and resource demands of deep learning methods limit their applicability in many practical scenarios. More unfortunately, the lack of interpretability of these models also poses a challenge to some application scenarios.

In contrast, the clear theoretical mechanism makes the shallow ML methods the other popular way for HSI classification. Moreover, the typical methods, such as support vector machines (SVMs) [12], random forests (RFs) [13], and linear regression (LR) [38], have been widely proven with low computational cost and resource requirements in many applications. Nevertheless, the high dimensionality and strong interband correlations inherent to HSIs exacerbate the so-called "curse of dimensionality," thereby presenting

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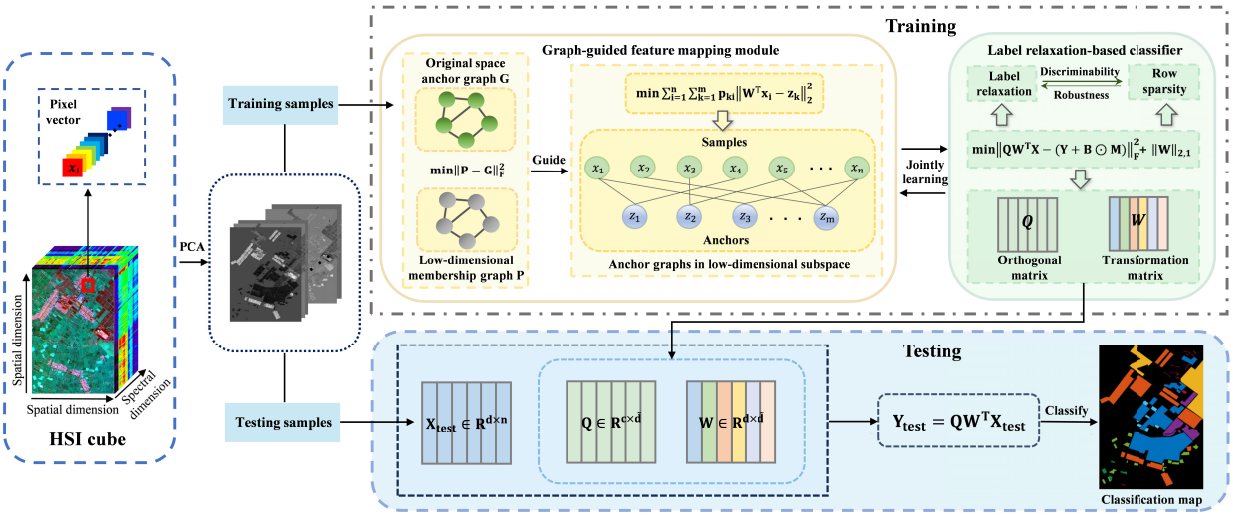


Fig. 1. Overall framework of the FAGRLR model for HSI classification.

more challenges for HSI classification using those methods directly. To address this issue, dimensionality reduction (DR) techniques [14] have been widely adopted. Traditional DR methods, including principal component analysis (PCA) [39], linear discriminant analysis (LDA) [40], and semi-supervised discriminant analysis (SDA) [41], aim to learn an optimal projection matrix that transforms high-dimensional data into a low-dimensional space. Advanced variants, such as bilateral PCA [42], 2D-PCA [43], kernel LDA [44], and regularized LDA [45], have extended these methods' capabilities. However, these methods primarily emphasize global structures and often neglect essential local geometric relationships. To address this limitation, DR methods based on low-rank representation (LRR) have been proposed. These methods exploit the inherent low-rank property of HSI data to provide a more adaptable representation. For example, discriminative low-rank projections (DLRPs) [46] integrate LRR with supervised learning to derive optimal low-rank subspaces and discriminative projections. Furthermore, LatLRE [47] combines latent LRR with projection learning for robust HSI DR. Alongside LRR-based techniques, manifold learning-based DR methods have gained significant attention. Representative methods include isometric feature mapping (ISOMAP) [48], locality preserving projections (LPPs) [49], and neighborhood preserving embedding (NPE) [50]. Recently, the spatial–spectral self-training and regularized metric constraint (SSMM) [51] method has been proposed to achieve more effective DR in HSI. Building on the principles of manifold learning, graph embedding has emerged as a general DR framework, exemplified by methods such as sparse and low-rank graph discriminant analysis (SLGDA) [52] and patch tensor-based sparse and low-rank graphs (PT-SLGs) [53]. To further enhance the preservation of local structures, tensor LPP (TLPP) and its improved version, modified TLPP (MTLPP) [54], have been proposed. The adaptive homophily structure graph clustering (AHSGC) [55] improves structural representation by adaptively refining graph topology, thereby implicitly transforming high-dimensional spectral features into discriminative low-dimensional embeddings while preserving spatial homophily. In parallel, the self-supervised efficient low-pass contrastive graph clustering (SLCGC) [56] employs homogeneous region aggregation and low-pass graph filtering to alleviate structural redundancy and reduce graph complexity, producing compact and noise-invariant latent representations. However, these methods often suffer from high computational costs due to the construction of adjacency matrices. To reduce this overhead, a fast anchor graph (AG) was introduced to reduce computational burden. In particular, a fast adaptive unsupervised projection model was constructed by using a few anchor points to build a bipartite graph for improving efficiency [57]. Fast AG preserving projections (FAGPPs) [58] replaced the adjacency matrix with a computationally efficient AG.

Although significant progress has been made in the above studies, LR-based methods typically rely on transforming training samples into strict binary-label matrices, which restricts model flexibility and hinders the learning of discriminative features. To overcome this limitation, Fang et al. [59] proposed the regularized label relaxation LR (RLR) model, introducing a nonnegative label relaxation matrix to soften the rigid binary labels. Building on this, Han et al. [60] developed the double relaxation regression (DRR) model, which simultaneously relaxes both the binary-label matrix and the graph regularization term using two transformation matrices. Further advances include Yang et al.'s [61] orthogonal autoencoder regression (OAR), which integrates RLR within an orthogonal autoencoder framework to enhance discriminative capability. RLR with spectral–spatial constrained locality adaptive regularization (SSLA-RLR) [62] and tensor decomposition-based relaxed LR (TDRLR) [63] have also been proposed. Additionally, Lai et al. [64] introduced a generalized robust regression (GRR) that decomposes the projection matrix into two separate components, thereby relaxing constraints on the number of projection classes. Other methods, including sparse representation classification (SRC) [65], collaborative representation classification (CRC) [66], discriminative least-squares regres-

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sion (DLSR) [67], principal space approximation ensemble discriminant edge least-squares regression (PSAE-DMLSR) [68], and dynamic label relaxation induced discriminative regression (DLRDR) [69], have further enriched this field. However, most of the aforementioned methods neglect the importance of regularization strategies for the transformation matrix. Traditional approaches often apply L_2-norm regularization to control the complexity of the transformation matrix. However, such uniform norm constraints often fail to fully exploit the intrinsic structure of hyperspectral data, limiting the ability of the low-dimensional space to represent critical class information while struggling to effectively suppress redundant information and noise. Although the L_1-norm encourages sparsity, it imposes independent constraints on individual elements of the transformation matrix, thereby neglecting the correlation among spectral bands, which may weaken the discriminative power of the reduced features. Consequently, enhancing the representational capacity of the model while effectively suppressing redundancy and noise remains a key challenge for HSI classification [70].

To address the above limitations, this article proposes a novel method called fast AG regularized RLR (FAGRLR) for HSI classification. The overall framework is depicted in Fig. 1. The proposed model comprises two main components: a graph-guided feature mapping module and a label-relaxed classifier module. In the feature mapping stage, an AG is first constructed in the original data space, followed by the formation of a membership graph in the reduced subspace. This design ensures consistency between the graph structures, effectively preserving local neighborhood information and yielding structure-preserving feature representations. Subsequently, during joint training, a label relaxation mechanism is introduced to mitigate the constraints of strict binary labels, thereby improving the model's discriminative capability and robustness against noise. Additionally, an $L_{2,1}$-norm regularization term is incorporated to promote sparsity, and the transformation matrix is jointly optimized to enable collaborative learning between the feature mapping and classifier components.

In summary, the main contributions of this article are as follows.

1) A novel classification model called FAGRLR is constructed by integrating fast AG into LR classifier, marking its first introduction and application in HSI classification. The fast AG approximates the global structure with a few anchor points, preserves topological relationships, and refines local manifolds while also reducing computational complexity.
2) Meanwhile, the label relaxation technique is introduced into the LR classifier to increase the label matrix's flexibility in discriminative transformation learning, thereby avoiding the risk of overfitting caused by using fixed labels. Furthermore, the $L_{2,1}$-norm constraint is imposed to learn a row sparse discriminative transformation matrix, which effectively mitigates the impact of outliers and significantly enhances the model's robustness.
3) To address the nonconvex optimization problem in the FAGRLR model, an iterative optimization algorithm is designed to allow the fast AG construction and classifier training to fine-tune each other during the iterations. Correspondingly, the convergence and computational complexity of the iterative algorithm are theoretically and numerically analyzed.

TABLE I
DETAILED VARIABLE DESCRIPTIONS IN THIS ARTICLE

Notation	Size	Description
$\mathbf{X}$	$d \times n$	data matrix
$\mathbf{Y}$	$c \times n$	label matrix
$\mathbf{G}$	$n \times m$	membership matrix
$\mathbf{P}$	$n \times m$	learned membership matrix
$\mathbf{A}$	$c \times d$	transformation matrix
$\mathbf{Z}$	$m \times d$	anchor graph matrix
$\mathbf{B}$	$c \times n$	luxury matrix
$\mathbf{M}$	$c \times n$	label relaxation matrix
$\mathbf{Q}$	$c \times \tilde{d}$	orthogonal matrix
$\mathbf{W}$	$d \times \tilde{d}$	transformation matrix

The remainder of this article is organized as follows. Section II presents the proposed model formulation and its optimization procedure. Section III reports the experimental results on four benchmark datasets for performance evaluation. Finally, Section IV concludes this article.

II. METHODOLOGY

A. Notations and Definitions

In HSI classification, the original HSI data cube of size $H \times W \times C$ is first unfolded into a 2-D spectral matrix, where each pixel in the spatial domain is reshaped into a C-dimensional spectral vector. Specifically, the 3-D cube is converted into a matrix $\mathbf{X} \in \mathbb{R}^{d \times n}$, where $n = H \times W$ denotes the total number of pixels and $d = C$ represents the number of spectral bands. Each column of $\mathbf{X}$ corresponds to the spectral signature of one pixel. In this article, we let the data matrix $\mathbf{X} = [x_1, x_2, \ldots, x_n]$, where $x_i \in \mathbb{R}^d$ and d is the feature dimension. The binary-label matrix is $\mathbf{Y} = [y_1, y_2, \ldots, y_n]$, where $y_i \in \mathbb{R}^c$ and c is the total number of classes. $\|\mathbf{A}\|_{2,1}$ denotes the $L_{2,1}$-norm of the matrix $\mathbf{A}$. Table I lists the detailed descriptions of all variables used in this article.

B. Formulation of FAGRLR

LR has been widely applied in HSI classification due to its simplicity and computational efficiency. However, most existing LR-based methods mainly focus on fitting the regression relationship between features and labels, while lacking the ability to explicitly model the intrinsic geometric structure of high-dimensional hyperspectral data. For HSIs, each pixel is represented by high-dimensional spectral features, and the relationships among neighboring samples are largely determined by spectral similarity. Without effectively preserving such local structural information, some important relationships among spectral features will be lost during feature projection,

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which can adversely affect the classification performance. To address this limitation, a fast AG is constructed in the spectral feature space and embedded into the LR framework to preserve the local spectral neighborhood structures during the feature projection. Accordingly, the objective function is formulated as follows:

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{P}} \quad & \|\mathbf{AX}-\mathbf{Y}\|_F^2+\alpha\|\mathbf{P}-\mathbf{G}\|_F^2 \\ & +\beta\sum_{i=1}^{n}\sum_{k=1}^{m}p_{ki}\left\|\mathbf{A}^\top\boldsymbol{x}_i-\boldsymbol{z}_k\right\|_2^2+\gamma\|\mathbf{A}\|_F^2 \\ \text{s.t.} \quad & \boldsymbol{p}_i^\top\mathbf{1}_m=1, \quad \mathbf{P}\geq 0 \end{aligned} \tag{1}$$

where $\mathbf{G}$ denotes the membership matrix computed in the original spectral feature space, $\mathbf{P}$ denotes the learned membership matrix, and z_k represents the kth anchor in the projected space. In particular, the term $\|\mathbf{P}-\mathbf{G}\|_F^2$ enforces consistency between the learned membership and the prior spectral-space membership, thereby preserving spectral neighborhood relationships among samples. Meanwhile, the weighted reconstruction term $\sum_{i=1}^{n}\sum_{k=1}^{m}p_{ki}\|\mathbf{A}^\top\boldsymbol{x}_i-\boldsymbol{z}_k\|_2^2$ encourages each projected spectral sample to remain close to its corresponding anchor under the learned membership weights, ensuring that spectral similarity is preserved in the low-dimensional subspace.

While the above model effectively preserves the local geometric structure of data, its reliance on rigid binary supervision imposes a fundamental limitation by enforcing strict mutual exclusivity among classes. This constraint ignores the semantic similarities among nontarget classes, particularly near ambiguous class boundaries, thereby reducing the model's discriminative capability. To overcome this limitation, a label relaxation strategy is adopted by introducing a relaxation variable matrix, which alleviates the strict binary constraints. Accordingly, the model can be rewritten as

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{M},\mathbf{P}} \quad & \|\mathbf{AX}-(\mathbf{Y}+\mathbf{B}\odot\mathbf{M})\|_F^2+\alpha\|\mathbf{P}-\mathbf{G}\|_F^2 \\ & +\beta\sum_{i=1}^{n}\sum_{k=1}^{m}p_{ki}\left\|\mathbf{A}^\top\boldsymbol{x}_i-\boldsymbol{z}_k\right\|_2^2+\gamma\|\mathbf{A}\|_F^2 \\ \text{s.t.} \quad & \mathbf{M}\geq 0, \quad \boldsymbol{p}_i^\top\mathbf{1}_m=1, \quad \mathbf{P}\geq 0 \end{aligned} \tag{2}$$

where $\odot$ denotes the Hadamard operator, $\mathbf{M}\in\mathbb{R}^{c\times n}$ is a nonnegative label relaxation matrix, and $\mathbf{B}\in\mathbb{R}^{c\times n}$ is a luxury matrix, which is defined as

$$\mathbf{B}_{ij}=\begin{cases}+1, & \text{if } \mathbf{Y}_{ij}=1\\ -1, & \text{if } \mathbf{Y}_{ij}=0.\end{cases} \tag{3}$$

Unfortunately, since RLR often suffers from the small-class problem, model (2) generally performs unsatisfactory when dealing with class-imbalanced datasets. To avoid this situation, the original transformation matrix $\mathbf{A}$ is decomposed into $\mathbf{QW}^\top$ to alleviate the constraint on the dimension of the projection space imposed by the number of classes [64], where $\mathbf{Q}\in\mathbb{R}^{c\times\tilde{d}}$ is an orthogonal matrix and $\mathbf{W}\in\mathbb{R}^{d\times\tilde{d}}$ is a transformation matrix (with $\tilde{d}$ denoting the dimension after DR). Thus, model (2) is converted to

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{W},\mathbf{Q},\mathbf{M},\mathbf{P}} \quad & \left\|\mathbf{QW}^\top\mathbf{X}-(\mathbf{Y}+\mathbf{B}\odot\mathbf{M})\right\|_F^2+\alpha\|\mathbf{P}-\mathbf{G}\|_F^2 \\ & +\beta\sum_{i=1}^{n}\sum_{k=1}^{m}p_{ki}\left\|\mathbf{W}^\top\boldsymbol{x}_i-\boldsymbol{z}_k\right\|_2^2+\gamma\|\mathbf{W}\|_F^2 \\ \text{s.t.} \quad & \mathbf{Q}^\top\mathbf{Q}=I, \quad \mathbf{M}\geq 0, \quad \boldsymbol{p}_i^\top\mathbf{1}_m=1, \quad \mathbf{P}\geq 0. \end{aligned} \tag{4}$$

In the above model, z_k denotes the kth anchor point in low-dimensional space transformed by $\mathbf{W}$, and the size of transformation matrix $\mathbf{W}$ is $d\times\tilde{d}$, which is not limited by the number of classes since $\tilde{d}$ can be set as any integer, Therefore, the small-class problem in model (2) can be effectively addressed.

However, the Frobenius norm has been widely proven to have insufficient robustness against noise and outliers, limiting the generalization ability of models in complex scenarios. To enhance the stability of the features learned by the above model, the $L_{2,1}$-norm is adopted to suppress irrelevant features and mitigate local perturbations. Therefore, the final formulation of the proposed FAGRLR model is expressed as follows:

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{W},\mathbf{Q},\mathbf{M},\mathbf{P}} \quad & \left\|\mathbf{QW}^\top\mathbf{X}-(\mathbf{Y}+\mathbf{B}\odot\mathbf{M})\right\|_F^2+\alpha\|\mathbf{P}-\mathbf{G}\|_F^2 \\ & +\beta\sum_{i=1}^{n}\sum_{k=1}^{m}p_{ki}\left\|\mathbf{W}^\top\boldsymbol{x}_i-\boldsymbol{z}_k\right\|_2^2+\gamma\|\mathbf{W}\|_{2,1} \\ \text{s.t.} \quad & \mathbf{Q}^\top\mathbf{Q}=I, \quad \mathbf{M}\geq 0, \quad \boldsymbol{p}_i^\top\mathbf{1}_m=1, \quad \mathbf{P}\geq 0. \end{aligned} \tag{5}$$

Notably, existing LR-based classification methods often struggle to adequately capture fine-grained local spectral relationships in high-dimensional data and can be sensitive to irrelevant or noisy bands, which reduces robustness and class separability. FAGRLR is designed to address these challenges by jointly integrating AG construction and $L_{2,1}$-norm regularization in a unified formulation. First, FAGRLR constructs the AG using balanced k-means-based hierarchical k-means (BKHK) to better preserve local spectral structure. This differs from the fast semi-supervised classification method based on AG (FSSCAG) [71], which mainly relies on neighbor-based assignments, and from the anchor-based graph embedding and soft label learning method for multilabel classification with missing labels (ASML) [72], which adopts a bipartite graph construction. Second, FAGRLR applies the $L_{2,1}$-norm directly to the projection matrix, which helps suppress uninformative or noisy features while retaining discriminative information. In contrast, methods such as the robust $L_{2,1}$-norm graph [73] and sparse regularized least-squares regression (LSR) [74] typically impose sparsity on coefficients or regression outputs rather than on the feature projection. By combining these two components, FAGRLR differs from prior AG and $L_{2,1}$-regularized approaches and yields more stable, discriminative feature representations for high-dimensional data analysis tasks.

C. Solving Algorithm

In this section, an optimization algorithm for FAGRLR is presented in detail. Since the objective function in (5) is not jointly convex with respect to $\mathbf{Z}$, $\mathbf{W}$, $\mathbf{Q}$, $\mathbf{M}$, and $\mathbf{P}$, an alternating iterative optimization algorithm is adopted to solve the problem efficiently.

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1) Update $\mathbf{Z}$*:* With $\mathbf{W}$, $\mathbf{Q}$, $\mathbf{M}$, and $\mathbf{P}$ being fixed, problem (5) can be transformed into the following problem with regard to $\mathbf{Z}$:

$$\min_{\mathbf{Z}} \beta \sum_{i=1}^{n}\sum_{k=1}^{m} p_{ki} \left\| \mathbf{W}^\top \boldsymbol{x}_i - \boldsymbol{z}_k \right\|_2^2. \tag{6}$$

The solution can be derived by setting the partial derivative of the objective function (6) with respect to z_k to zero, which yields the following expression:

$$\sum_{i=1}^{n} \left(p_{ki}\mathbf{W}^\top \boldsymbol{x}_i - p_{ki}\boldsymbol{z}_k \right) = 0. \tag{7}$$

Thus, the optimal $\mathbf{Z}$ can be updated by

$$\boldsymbol{z}_k = \frac{\mathbf{W}^\top \sum_{i=1}^{n} p_{ki}\boldsymbol{x}_i}{\sum_{i=1}^{n} p_{ki}}. \tag{8}$$

2) Update $\mathbf{W}$*:* Similarly, problem (5) with respect to $\mathbf{W}$ can be formulated as follows:

$$\begin{aligned} \min_{\mathbf{W}} \ & \left\| \mathbf{Q}\mathbf{W}^\top \mathbf{X} - (\mathbf{Y} + \mathbf{B} \odot \mathbf{M}) \right\|_F^2 \\ & + \beta \sum_{i=1}^{n}\sum_{k=1}^{m} p_{ki} \left\| \mathbf{W}^\top \boldsymbol{x}_i - \boldsymbol{z}_k \right\|_2^2 + \gamma \left\| \mathbf{W} \right\|_{2,1}. \end{aligned} \tag{9}$$

Denote $\boldsymbol{f}_k = \left(\sum_{i=1}^{n} p_{ki}\boldsymbol{x}_i\right) / \left(\sum_{i=1}^{n} p_{ki}\right)$, and we can obtain $\boldsymbol{z}_k = \mathbf{W}^\top \boldsymbol{f}_k$. Then, the above problem can be reformulated as follows:

$$\begin{aligned} \min_{\mathbf{W}} \ & \left\| \mathbf{Q}\mathbf{W}^\top \mathbf{X} - (\mathbf{Y} + \mathbf{B} \odot \mathbf{M}) \right\|_F^2 \\ & + \beta \sum_{i=1}^{n}\sum_{k=1}^{m} p_{ki} \left\| \mathbf{W}^\top \boldsymbol{x}_i - \mathbf{W}^\top \boldsymbol{f}_k \right\|_2^2 + \gamma \left\| \mathbf{W} \right\|_{2,1}. \end{aligned} \tag{10}$$

Let $\mathbf{K} = \mathbf{Y} + \mathbf{B} \odot \mathbf{M}$. The first term in model (10) can be transformed into

$$\begin{aligned} & \left\| \mathbf{Q}\mathbf{W}^\top \mathbf{X} - (\mathbf{Y} + \mathbf{B} \odot \mathbf{M}) \right\|_F^2 \\ & = \mathrm{tr}\left(\left(\mathbf{Q}\mathbf{W}^\top \mathbf{X} - \mathbf{K} \right)^\top \left(\mathbf{Q}\mathbf{W}^\top \mathbf{X} - \mathbf{K} \right) \right). \end{aligned} \tag{11}$$

Following the method in [58], the second term in model (10) can be rewritten as:

$$\sum_{i=1}^{n}\sum_{k=1}^{m} p_{ki} \left\| \mathbf{W}^\top \boldsymbol{x}_i - \mathbf{W}^\top \boldsymbol{f}_k \right\|_2^2 = \mathrm{tr}\left(\mathbf{W}^\top \tilde{\mathbf{G}} \mathbf{W} \right) \tag{12}$$

where $\mathbf{G} = \mathbf{X}^\top \mathbf{A}^{\mathrm{row}} \mathbf{X} + \mathbf{Z}^\top \mathbf{A}^{\mathrm{col}} \mathbf{Z} - \mathbf{X}^\top \mathbf{P}\mathbf{Z} - (\mathbf{X}^\top \mathbf{P}\mathbf{Z})^\top$ and $\tilde{\mathbf{G}} = (\mathbf{G} + \mathbf{G}^\top)/2$. $\mathbf{A}^{\mathrm{row}} \in \mathbb{R}^{n \times n}$ and $\mathbf{A}^{\mathrm{col}} \in \mathbb{R}^{c \times c}$ are two diagonal matrices. Specifically, the ith diagonal element of $\mathbf{A}^{\mathrm{row}}$ is defined as the sum of the elements in the ith row of matrix $\mathbf{P}$, i.e., $\mathbf{A}_{ii}^{\mathrm{row}} = \sum_{j=1}^{c} p_{ij}$; similarly, the jth diagonal element of $\mathbf{A}^{\mathrm{col}}$ corresponds to the sum of the elements in the jth column, i.e., $\mathbf{A}_{jj}^{\mathrm{col}} = \sum_{i=1}^{n} p_{ij}$.

We further introduce a diagonal matrix $\mathbf{D}$, defined as

$$\mathbf{D}_{ii} = \frac{1}{2\left\| \boldsymbol{w}^i \right\|_2} \tag{13}$$

where $\boldsymbol{w}^i$ denotes the ith row of $\mathbf{W}$. Then, the minimization of the $L_{2,1}$-norm in model (10) can be equivalently expressed as the following trace-form minimization problem:

$$\min_{\mathbf{W}} \left\| \mathbf{W} \right\|_{2,1} = \min_{\mathbf{W}} \mathrm{tr}\left(\mathbf{W}\mathbf{D}\mathbf{W}^\top \right). \tag{14}$$

Putting (11), (12), and (14) together, model (15) can be equivalently reformulated as

$$\begin{aligned} \min_{\mathbf{W}} \ & \mathrm{tr}\left(\left(\mathbf{Q}\mathbf{W}^\top \mathbf{X} - \mathbf{K} \right)^\top \left(\mathbf{Q}\mathbf{W}^\top \mathbf{X} - \mathbf{K} \right) \right) \\ & + \mathrm{tr}\left(\mathbf{W}^\top \tilde{\mathbf{G}} \mathbf{W} \right) + \mathrm{tr}\left(\mathbf{W}^\top \mathbf{D}\mathbf{W} \right). \end{aligned} \tag{15}$$

Then, the optimal $\mathbf{W}$ can be updated by

$$\begin{aligned} & \left(\mathbf{X}\mathbf{X}^\top + \beta\tilde{\mathbf{G}} + \gamma D \right) \mathbf{W} = \mathbf{X}\mathbf{K}^\top \mathbf{Q} \\ & \Rightarrow \mathbf{W} = \left(\mathbf{X}\mathbf{X}^\top + \beta\tilde{\mathbf{G}} + \gamma D \right)^{-1} \mathbf{X}\mathbf{K}^\top \mathbf{Q}. \end{aligned} \tag{16}$$

3) Update $\mathbf{Q}$*:* While other variables are fixed, the minimization problem (5) with respect to $\mathbf{Q}$ is transformed into

$$\begin{aligned} \min_{\mathbf{Q}} \ & \left\| \mathbf{Q}\mathbf{W}^\top \mathbf{X} - \mathbf{K} \right\|_F^2 \\ & \text{s.t. } \mathbf{Q}^\top \mathbf{Q} = I. \end{aligned} \tag{17}$$

Then, the optimal $\mathbf{Q}$ can be obtained by solving the following equivalent maximization problem:

$$\begin{aligned} \max_{\mathbf{Q}} \ & \mathrm{tr}\left(\mathbf{Q}^\top \mathbf{K}\mathbf{X}^\top \mathbf{W} \right) \\ & \text{s.t. } \mathbf{Q}^\top \mathbf{Q} = I. \end{aligned} \tag{18}$$

The optimal $\mathbf{Q}$ is updated by [75]

$$\mathbf{Q} = \mathbf{U}\mathbf{V}^\top. \tag{19}$$

Here, $\mathbf{U}\mathbf{S}\mathbf{V}^\top$ = SVD $(\mathbf{K}\mathbf{X}^\top\mathbf{W})$ and SVD $(\cdot)$ represents the singular value decomposition (SVD) of a matrix.

4) Update $\mathbf{M}$*:* By fixing all variables except $\mathbf{M}$, model (5) can be transformed into the following problem:

$$\begin{aligned} \min_{\mathbf{M}} \ & \left\| \mathbf{F} - \mathbf{B} \odot \mathbf{M} \right\|_F^2 \\ & \text{s.t. } \mathbf{M} \geq 0 \end{aligned} \tag{20}$$

where $\mathbf{F} = \mathbf{Q}\mathbf{W}^\top\mathbf{X} - \mathbf{Y}$. Referring to [62], the optimal solution of $\mathbf{M}$ is updated by the following equation:

$$\mathbf{M} = \max\left(\mathbf{B} \odot \mathbf{F}, 0 \right). \tag{21}$$

5) Update $\mathbf{P}$*:* With variables $\mathbf{W}$, $\mathbf{Q}$, $\mathbf{Z}$, and $\mathbf{M}$ being fixed, problem (5) can be simplified into the following problem:

$$\begin{aligned} \min_{\mathbf{P}} \ & \alpha \left\| \mathbf{P} - \mathbf{G} \right\|_F^2 + \beta \sum_{i=1}^{n}\sum_{k=1}^{m} p_{ki} \left\| \mathbf{W}^\top \boldsymbol{x}_i - \boldsymbol{z}_k \right\|_2^2 \\ & \text{s.t. } \mathbf{P} \geq 0, \ \boldsymbol{p}_i^\top \mathbf{1}_m = 1. \end{aligned} \tag{22}$$

Leaving $d_{ki} = \left\| \mathbf{W}^\top \boldsymbol{x}_i - \boldsymbol{z}_k \right\|_2^2$, (22) is transformed into

$$\begin{aligned} \min_{\mathbf{P}} \ & \alpha \left\| \mathbf{P} - \mathbf{G} \right\|_F^2 + \beta \sum_{i=1}^{n}\sum_{k=1}^{m} p_{ki} d_{ki} \\ & \text{s.t. } \mathbf{P} \geq 0, \ \boldsymbol{p}_i^\top \mathbf{1}_m = 1. \end{aligned} \tag{23}$$

Since the variable $\boldsymbol{p}_i$ is subject to both linear and nonnegativity constraints simultaneously, the above problem can be reformulated as

$$\begin{aligned} \min_{\boldsymbol{p}_i} \ & \frac{1}{2} \left\| \boldsymbol{p}_i - \boldsymbol{h}_i \right\|_2^2 \\ & \text{s.t. } \boldsymbol{p}_i^\top \mathbf{1}_m = 1, \ \boldsymbol{p}_i \geq 0 \end{aligned} \tag{24}$$

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TABLE II
COMPUTATIONAL COST OF DIFFERENT METHODS ON FOUR DATASETS

Methods	Indian Pines		Salinas		HeiHe		WHU-Hi-HanChuan	
	Tr(s)	Te(s)	Tr(s)	Te(s)	Tr(s)	Te(s)	Tr(s)	Te(s)
SVM	0.916	11.421	0.201	7.959	0.298	9.003	1.112	111.356
RLR	0.375	0.003	**0.103**	0.008	**0.195**	0.007	0.305	0.126
DLRP	87.757	0.184	21.023	0.468	82.953	0.668	7.881	2.239
DRR	0.226	0.006	0.226	0.006	0.680	0.007	0.225	0.029
OAR	1.073	0.002	1.231	0.023	0.915	0.008	0.645	0.049
SSLA-RLR	0.414	0.010	0.159	0.009	0.247	0.009	0.535	0.183
PSAE-DMLSR	45.415	26.526	65.564	32.617	88.261	52.273	54.713	32.169
TDRLR	13.176	5.732	12.152	3.165	14.716	4.275	20.198	8.131
LatLRE	8.627	1.716	5.266	1.290	7.619	0.942	15.193	2.550
FAGRLR	**0.257**	**0.001**	0.145	**0.005**	0.271	**0.006**	**0.114**	**0.023**

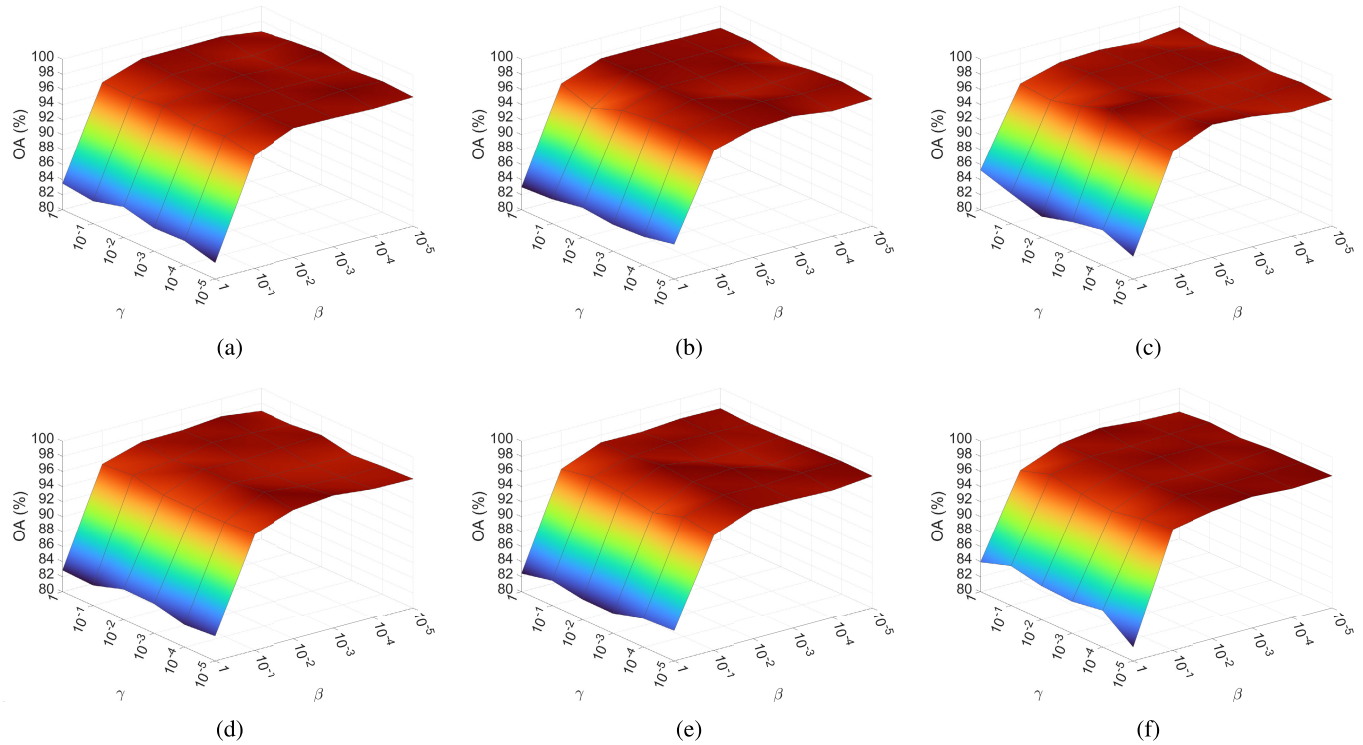


Fig. 2. Variation of OA with β and γ when α is set to different values for the Indian Pines dataset. (a) $\alpha = 1$. (b) $\alpha = 0.1$. (c) $\alpha = 0.01$. (d) $\alpha = 0.001$. (e) $\alpha = 0.0001$. (f) $\alpha = 0.00001$.

TABLE III
COMPARISON OF DIFFERENT DR METHODS ON INDIAN PINES DATASET

Method	OA (%)
LPP	94.31
NPE	92.98
PT-SLG	98.86
PCA	96.69

where $\boldsymbol{h}_i = ((g_{1i} - \beta d_{1i})/2\alpha, \ldots, (g_{mi} - \beta d_{mi})/2\alpha)^\top$. Thus, the corresponding augmented Lagrangian function can be formulated as follows:

$$\frac{1}{2}\left\|\boldsymbol{p}_i - \boldsymbol{h}_i\right\|_2^2 - \boldsymbol{\eta}_i\left(\boldsymbol{p}_i^\top \mathbf{1}_m - 1\right) - \boldsymbol{\theta}_i^\top \boldsymbol{p}_i \tag{25}$$

where $\boldsymbol{\eta}_i$ and $\boldsymbol{\theta}_i$ are two Lagrangian multipliers.

Let the optimal solution to (25) be denoted as $\boldsymbol{p}_i^*$, and the corresponding optimal values of $\boldsymbol{\eta}_i$ and $\boldsymbol{\theta}_i$ be $\boldsymbol{\eta}_i^*$ and $\boldsymbol{\theta}_i^*$, respectively. Denote $\bar{\boldsymbol{\theta}}_i^* = (\mathbf{1}_m^\top \boldsymbol{\theta}_i^*)/m$ and $\boldsymbol{\xi}_i = \boldsymbol{h}_i - (\mathbf{1}_m)/m\boldsymbol{h}_i\mathbf{1}_m + 1/m\mathbf{1}_m$. Using the Karush–Kuhn–Tucker (KKT) conditions, $\boldsymbol{p}_i^*$ can be updated as follows:

$$\boldsymbol{p}_{ki}^* = \left(\boldsymbol{\xi}_{ki} - \bar{\boldsymbol{\theta}}_i^*\right)_+ \tag{26}$$

where $\boldsymbol{p}_{ki}^*$ and $\boldsymbol{\xi}_{ki}$ are the kth element of $\boldsymbol{p}_i^*$ and $\boldsymbol{\xi}_i$, respectively.

D. Convergence Analysis

In this section, the convergence of the FAGRLR algorithm is analyzed. Let the value of objective function in (5) be

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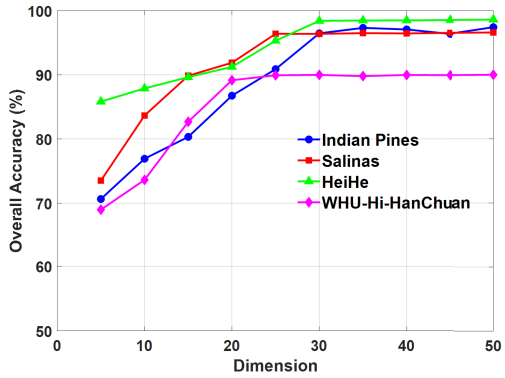


Fig. 3. Sensitivity analysis for the proposed method with different dimensions on four HSI datasets.

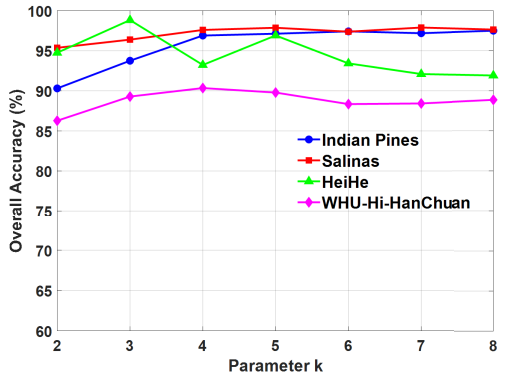


Fig. 4. Sensitivity analysis for the proposed method with respect to the parameter k on four HSI datasets.

computed as

$$\begin{aligned}&\mathbf{J}(\mathbf{Z},\mathbf{W},\mathbf{Q},\mathbf{M},\mathbf{P})\\&=\left\|\mathbf{Q}\mathbf{W}^{\top}\mathbf{X}-(\mathbf{Y}+\mathbf{B}\odot\mathbf{M})\right\|_F^2+\alpha\|\mathbf{P}-\mathbf{G}\|_F^2\\&\quad+\beta\sum_{i=1}^{n}\sum_{k=1}^{m}p_{ki}\left\|\mathbf{W}^{\top}\boldsymbol{x}_i-\boldsymbol{z}_k\right\|_2^2+\gamma\|\mathbf{W}\|_{2,1}.\end{aligned}\tag{27}$$

Without loss of generality, assume that the solution of (5) is given by $\mathbf{W_t}, \mathbf{Q_t}, \mathbf{M_t}, \mathbf{Z_t}$ and $\mathbf{P_t}$ at the tth iteration. According to (6), when $\mathbf{W}, \mathbf{Q}, \mathbf{M}$, and $\mathbf{P}$ are fixed, the optimal value of $\mathbf{Z}$ is obtained by taking the partial derivative of (5) with respect to $\mathbf{Z}$ and setting it to zero. Thus, the following inequality can be derived:

$$\mathbf{J}\left(\mathbf{W_t},\mathbf{Q_t},\mathbf{M_t},\mathbf{Z_{t+1}},\mathbf{P_t}\right)\le\mathbf{J}\left(\mathbf{W_t},\mathbf{Q_t},\mathbf{M_t},\mathbf{Z_t},\mathbf{P_t}\right).\tag{28}$$

Next, by reformulating (9) in terms of the matrix's trace, the optimal solution for $\mathbf{W}$ is also obtained by setting its partial derivative to zero. This leads to the following inequality:

$$\mathbf{J}\left(\mathbf{W_{t+1}},\mathbf{Q_t},\mathbf{M_t},\mathbf{Z_{t+1}},\mathbf{P_t}\right)\le\mathbf{J}\left(\mathbf{W_t},\mathbf{Q_t},\mathbf{M_t},\mathbf{Z_{t+1}},\mathbf{P_t}\right).\tag{29}$$

Since $\mathbf{Q}$ is subject to the orthogonality constraint $\mathbf{Q}^{\top}\mathbf{Q}=I$, the optimal solution of $\mathbf{Q}$ is calculated based on the SVD of $\mathbf{K}\mathbf{X}^{\top}\mathbf{W}$ by referring to [75]. Therefore, the following inequality holds:

$$\mathbf{J}\left(\mathbf{W_{t+1}},\mathbf{Q_{t+1}},\mathbf{M_t},\mathbf{Z_{t+1}},\mathbf{P_t}\right)\le\mathbf{J}\left(\mathbf{W_{t+1}},\mathbf{Q_t},\mathbf{M_t},\mathbf{Z_{t+1}},\mathbf{P_t}\right).\tag{30}$$

For $\mathbf{M}$, its optimal solution can be obtained from (21). Then, we have

$$\begin{aligned}&\mathbf{J}\left(\mathbf{W_{t+1}},\mathbf{Q_{t+1}},\mathbf{M_{t+1}},\mathbf{Z_{t+1}},\mathbf{P_t}\right)\\&\le\mathbf{J}\left(\mathbf{W_{t+1}},\mathbf{Q_{t+1}},\mathbf{M_t},\mathbf{Z_{t+1}},\mathbf{P_t}\right).\end{aligned}\tag{31}$$

In the next step, the updated $\mathbf{W}, \mathbf{Q}, \mathbf{Z}$, and $\mathbf{M}$ are used to estimate $\mathbf{P}$. Therefore,

$$\begin{aligned}&\mathbf{J}\left(\mathbf{W_{t+1}},\mathbf{Q_{t+1}},\mathbf{M_{t+1}},\mathbf{Z_{t+1}},\mathbf{P_{t+1}}\right)\\&\le\mathbf{J}\left(\mathbf{W_{t+1}},\mathbf{Q_{t+1}},\mathbf{M_{t+1}},\mathbf{Z_{t+1}},\mathbf{P_t}\right).\end{aligned}\tag{32}$$

Combining inequalities (28)–(32) together, the following inequality can be derived:

$$\begin{aligned}&\mathbf{J}\left(\mathbf{W_{t+1}},\mathbf{Q_{t+1}},\mathbf{M_{t+1}},\mathbf{Z_{t+1}},\mathbf{P_{t+1}}\right)\\&\le\mathbf{J}\left(\mathbf{W_t},\mathbf{Q_t},\mathbf{M_t},\mathbf{Z_t},\mathbf{P_t}\right).\end{aligned}\tag{33}$$

The above inequality clearly indicates that the objective function value decreases from step t to step $t+1$. Furthermore, based on the properties of matrix norms, the objective function value $\mathbf{J}(\mathbf{Z},\mathbf{W},\mathbf{Q},\mathbf{M},\mathbf{P})$ is nonnegative. Therefore, it can be concluded that the objective function value decreases monotonically with each iteration and has a lower bound. This suggests that the iterative procedure outlined in Algorithm 1 is convergent.

Algorithm 1 Optimization of FAGRLR

Input: Data matrix $\mathbf{X}\in\mathbb{R}^{d\times n}$, label matrix $\mathbf{Y}\in\mathbb{R}^{c\times n}$, the luxury matrix $\mathbf{B}\in\mathbb{R}^{c\times n}$, number of anchor points m, reduced dimension $\tilde{d}$, and trade-off parameters α,β,γ.
Initialization: Non-negative label matrix $\mathbf{M}\in\mathbb{R}^{c\times n}$, orthogonal matrix $\mathbf{Q}\in\mathbb{R}^{c\times\tilde{d}}$, diagonal matrix $\mathbf{D}\in\mathbb{R}^{\tilde{d}\times\tilde{d}}$.
Steps:
Calculate membership matrix $\mathbf{G}$ using BKHK in the original space.
Assign initial value to matrix $\mathbf{P}$: $\mathbf{P}\leftarrow\mathbf{G}$.
while not converged **do**
  Update $\mathbf{Z}$ by solving (8).
  Update $\mathbf{W}$ by solving (16).
  Update $\mathbf{Q}$ by solving (19).
  Update $\mathbf{M}$ by solving (21).
  Update $\mathbf{P}$ by solving (25).
end while
Output: Orthogonal matrix $\mathbf{Q}$ and transformation matrix $\mathbf{W}$.

E. Computational Complexity

Consider a data matrix $\mathbf{X}\in\mathbb{R}^{d\times n}$ consisting of n samples with d-dimensional features. Initially, the proposed FAGRLR algorithm employs BKHK to generate the anchor-based membership matrix $\mathbf{G}$ from the original data. This step has a time complexity of $O(nd\log(m)t)$, where m denotes the number of anchors and t is a constant related to the number of clustering

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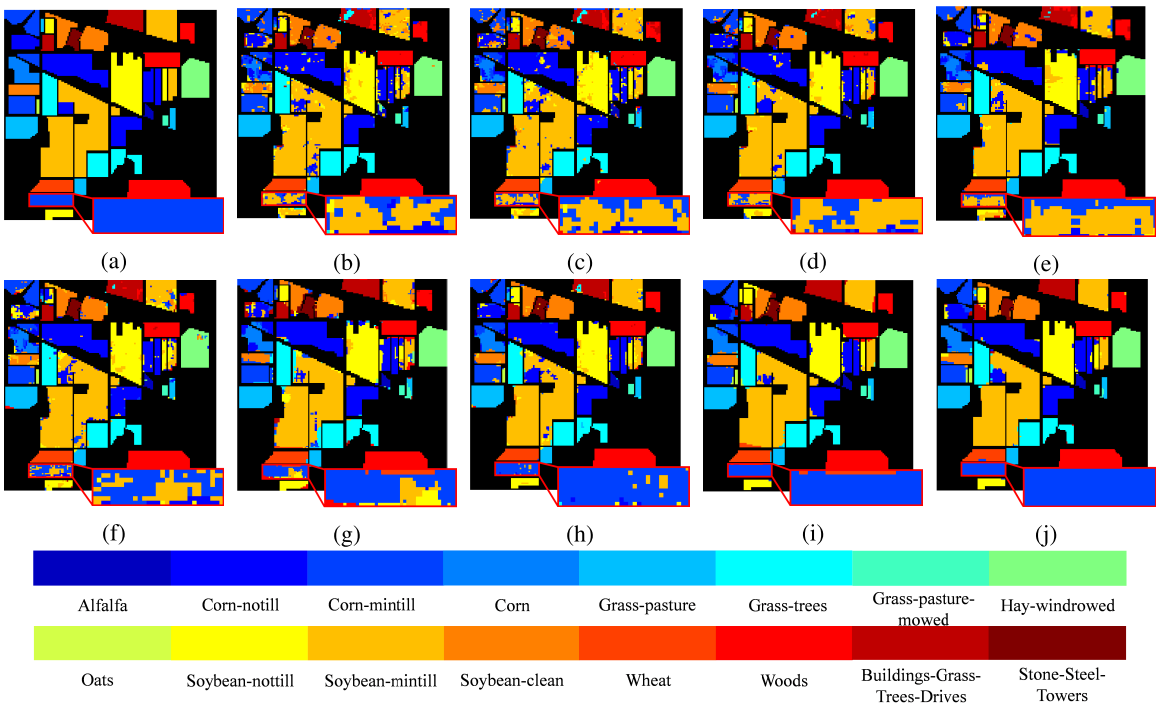


Fig. 5. Classification maps of different methods for Indian Pines dataset. (a) Original, (b) RLR, (c) DLRP, (d) DRR, (e) OAR, (f) SSLA-RLR, (g) PSAE-DMLSR, (h) TDRLR, (i) LatLRE, and (j) **FAGRLR**.

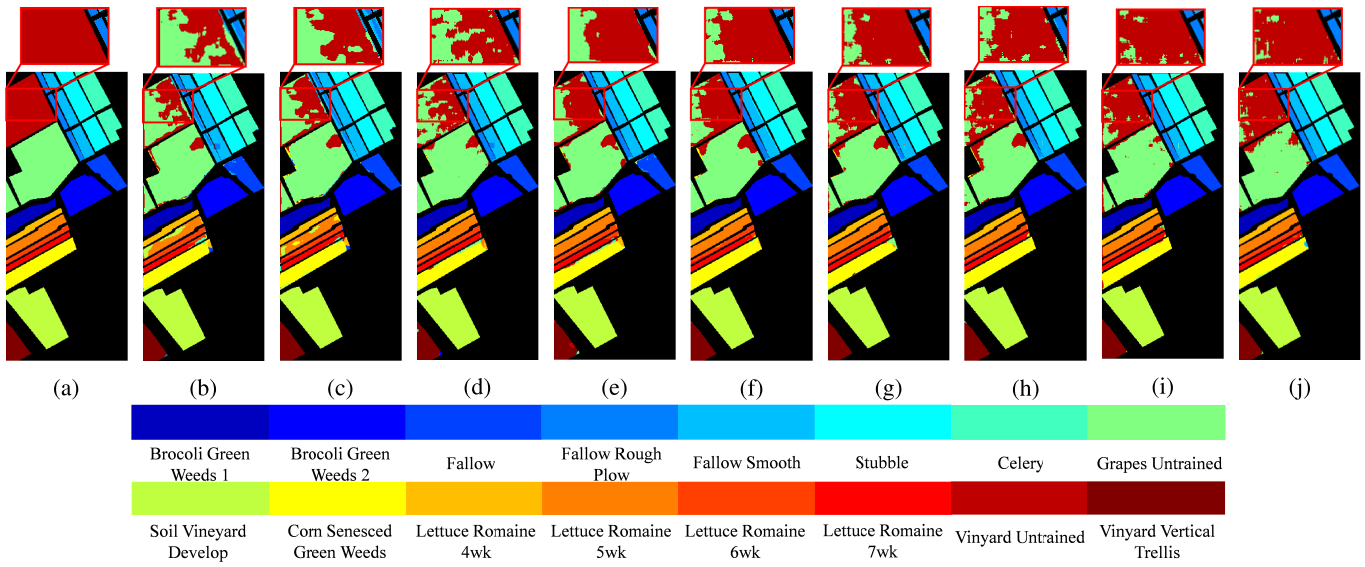


Fig. 6. Classification maps of different methods for the Salinas dataset. (a) Original, (b) RLR, (c) DLRP, (d) DRR, (e) OAR, (f) SSLA-RLR, (g) PSAE-DMLSR, (h) TDRLR, (i) LatLRE, and (j) **FAGRLR**.

layers. In each iteration of the subsequent optimization phase, the variables $\mathbf{Z}, \mathbf{W}, \mathbf{Q}, \mathbf{M}$, and $\mathbf{P}$ are sequentially updated. Updating $\mathbf{Z}$ incurs a computational complexity of $O(nmd)$. The optimization of $\mathbf{W}$ mainly involves matrix multiplication and inversion, resulting in a complexity of $O(nd^2 + d^3)$. Updating $\mathbf{Q}$ requires performing the SVD of $\mathbf{K}\mathbf{X}^\top\mathbf{W}$, with a cost of at most $O(cm\tilde{d})$ operations, where $\tilde{d}$ denotes the reduced feature dimension. Updating $\mathbf{M}$ has a complexity of $O(cn)$, reflecting label-level reconstruction across c classes and n samples. Finally, the update of matrix $\mathbf{P}$ requires $O(nmd)$ operations. Combining the above components, the overall computational complexity of each iteration of FAGRLR is approximately $O(nmd + nd^2 + d^3 + cm\tilde{d} + cn + nmd)$. For large datasets, where $m \ll n$, $c \ll n$, $d \ll n$, and $\tilde{d} \ll n$, the time complexity of the proposed algorithm is $O(nd^2)$ in each iteration. Furthermore, assuming that the algorithm converges after T iterations, the total computational complexity of FAGRLR is $O(Tnd^2)$, where T denotes the total number of iterations until convergence. Table II summarizes thecomputational cost comparisons of the different methods on four datasets. Tr represents the training time, while Te represents the testing time. For clarity, the best results belonging to our own

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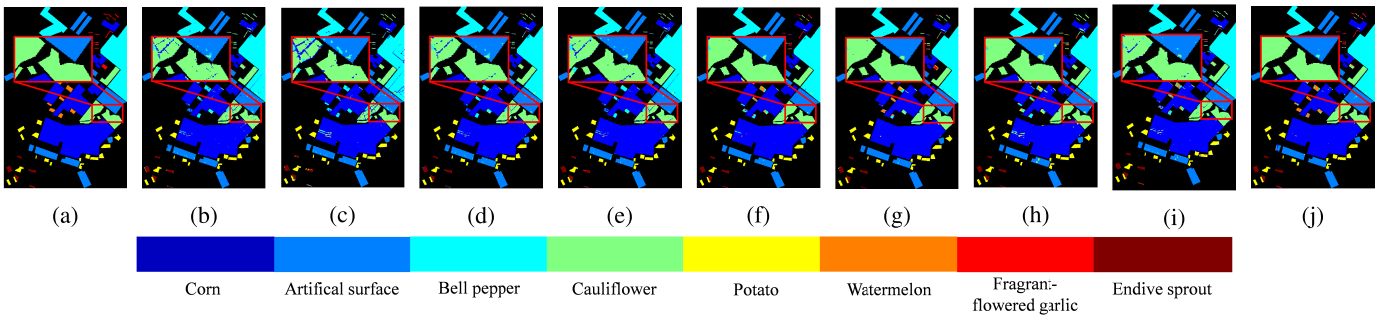


Fig. 7. Classification maps of different methods for the HeiHe dataset. (a) Original, (b) RLR, (c) DLRP, (d) DRR, (e) OAR, (f) SSLA-RLR, (g) PSAE-DMLSR, (h) TDRLR, (i) LatLRE, and (j) **FAGRLR**.

method are highlighted in bold when compared with other methods.

III. Experiments

This section begins with a detailed presentation of four HSI datasets employed in the experiments, including the Indian Pines, Salinas, HeiHe, and WHU-Hi-HanChuan datasets. Following this, a series of comparative experiments is conducted to evaluate the performance of the proposed method against several state-of-the-art algorithms. The section concludes with a thorough analysis and discussion of the experimental results, presented both quantitatively and qualitatively.

A. Datasets Description

The first dataset was collected by the AVIRIS in 1992, covering an agricultural area in northwestern Indiana, USA. It has a spatial size of 145×145 pixels and originally contained 224 spectral bands. After discarding four zero-value bands and 20 bands with low signal-to-noise ratios due to atmospheric absorption, 200 effective channels were retained. This dataset consists of 16 land-cover categories, with a total of 10 366 labeled pixel samples, most of which correspond to various crop types. For this dataset, 10% of the samples are used for training.

The second dataset was collected by the AVIRIS in 1998 in the Salinas Valley region of southern California, USA, with a spatial resolution of 3.7 m/pixel. It has a spatial dimension of 512×217 pixels, with the original data containing 224 spectral bands. After removing 20 bands affected by water absorption, 204 effective channels were retained. The dataset includes 16 land-cover classes, primarily consisting of crops, bare soil, and vineyards. For this dataset, 1% of labeled samples are used for training.

The third dataset was captured by the CASI/SASI sensor in 2012, within the Zhangye Basin of the central Heihe River Basin in Gansu Province, China. It has a spatial resolution of 684×453 pixels and includes eight distinct land-cover categories. To improve data quality, 14 spectral bands heavily affected by noise were removed, leaving 135 effective spectral bands for analysis. For this dataset, 1% of the labeled samples were used for training.

The fourth dataset was collected in 2016 from Hanchuan City in Hubei Province, China. It has an image resolution of 1217×303 pixels and contains 274 spectral bands, covering wavelengths ranging from 400 to 1000 nm. This dataset includes 17 distinct land-cover categories, such as strawberry, cowpea, water spinach, and various leafy vegetables. For this dataset, 20 samples from each class were selected as the training data.

B. Experimental Setup

To demonstrate the effectiveness of the proposed algorithm, we compared it with a range of recent and representative approaches. These include several ML-based methods, namely, RLR [59], DLRP [46], DRR [60], OAR [61], SSLA-RLR [62], PSAE-DMLSR [68], TDRLR [63], and LatLRE [47]. We also considered a number of DL-based methods for comparison, including WFCG [23], GPF-NET [24], GraphGST [27], 3DSS-Mamba [32], DiffFormer [29], and MambaHSI+ [33]. The accuracy of each class (EA), overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) were used as evaluation metrics to quantitatively assess the performance of different methods. To reduce the impact of sample bias, each method was executed ten times, and the results are reported as both the average values and the standard deviations. To prevent singularity issues, PCA was applied to preprocess the data. Table III presents the classification results on the Indian Pines dataset using different preprocessing methods, including LPP [49], NPE [50], PT-SLG [53], and PCA [39]. Although PT-SLG demonstrates superior classification accuracy in certain scenarios, it typically introduces additional hyperparameters and relies on specific prior assumptions and graph construction strategies. These factors significantly increase the overall model complexity and the burden of parameter tuning. Given that the primary objective of this study is to objectively evaluate the effectiveness of the proposed classification model, rather than optimizing the DR stage, we have chosen PCA as a more appropriate DR strategy to ensure fairness in the experiments.

The FAGRLR involves three key regularization parameters, namely, α, β, and γ. To determine the optimal parameter configuration, a fixed-parameter tuning strategy was adopted. Specifically, a grid search approach was employed, with each regularization parameter systematically explored within the set $\{0.00001, 0.0001, 0.001, 0.01, 0.1, 1\}$. Fig. 2 illustrates the variation of OA with respect to β and γ under different values

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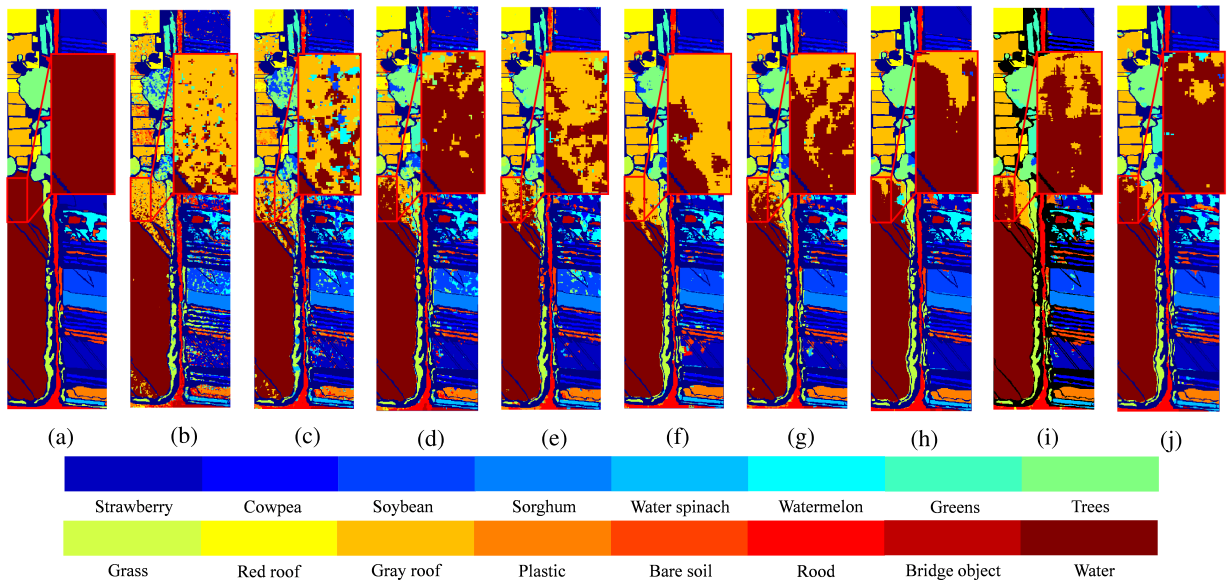


Fig. 8. Classification maps of different methods for the WHU-Hi-HanChuan dataset. (a) Original, (b) RLR, (c) DLRP, (d) DRR, (e) OAR, (f) SSLA-RLR, (g) PSAE-DMLSR, (h) TDRLR, (i) LatLRE, and (j) **FAGRLR**.

TABLE IV
CLASSIFICATION ACCURACY (%) OF INDIAN PINES DATASET WITH DIFFERENT CLASSIFICATION METHODS

Class	RLR	DLRP	DRR	OAR	SSLA-RLR	PSAE-DMLSR	TDRLR	LatLRE	**FAGRLR**
1	93.75 ± 3.49	92.25 ± 6.68	89.58 ± 7.97	95.01 ± 3.51	95.83 ± 4.39	96.11 ± 2.98	96.89 ± 3.68	55.71 ± 3.16	$\mathbf{97.77 \pm 2.55}$
2	75.89 ± 5.85	81.91 ± 3.07	88.76 ± 1.22	92.21 ± 1.93	94.74 ± 2.64	96.23 ± 0.87	96.37 ± 0.85	$\mathbf{97.64 \pm 0.14}$	96.07 ± 0.78
3	85.14 ± 2.94	66.23 ± 1.70	74.96 ± 3.22	65.90 ± 5.29	90.85 ± 1.99	93.93 ± 2.31	91.34 ± 1.70	94.57 ± 2.24	$\mathbf{96.18 \pm 1.47}$
4	83.17 ± 4.40	44.29 ± 4.01	63.55 ± 6.48	50.00 ± 7.65	83.92 ± 5.53	89.57 ± 3.95	90.09 ± 4.59	$\mathbf{98.06 \pm 0.75}$	94.29 ± 2.85
5	93.33 ± 3.69	95.90 ± 1.09	92.70 ± 3.31	92.22 ± 2.66	95.83 ± 1.99	96.06 ± 2.32	$\mathbf{97.52 \pm 1.59}$	92.15 ± 1.98	97.31 ± 1.14
6	97.98 ± 2.26	$\mathbf{99.72 \pm 0.53}$	98.51 ± 0.65	99.61 ± 0.79	99.25 ± 0.85	99.42 ± 0.49	98.23 ± 1.76	98.84 ± 2.02	99.16 ± 0.59
7	91.82 ± 3.56	93.74 ± 1.56	98.61 ± 2.77	95.68 ± 4.95	98.81 ± 3.77	99.72 ± 1.69	98.61 ± 2.77	85.96 ± 2.16	$\mathbf{99.90 \pm 0.21}$
8	99.76 ± 0.32	98.12 ± 0.98	99.53 ± 0.37	99.88 ± 0.12	$\mathbf{99.95 \pm 0.14}$	99.81 ± 0.26	99.30 ± 0.51	99.15 ± 1.16	99.81 ± 0.32
9	72.87 ± 2.25	78.55 ± 4.32	90.72 ± 5.88	41.00 ± 11.00	99.00 ± 3.16	98.00 ± 4.21	98.40 ± 3.72	46.71 ± 5.76	$\mathbf{99.95 \pm 0.47}$
10	73.62 ± 1.82	72.38 ± 1.92	75.65 ± 4.73	81.25 ± 4.18	90.99 ± 3.71	94.21 ± 0.99	95.06 ± 0.80	88.36 ± 3.64	$\mathbf{95.34 \pm 1.15}$
11	89.57 ± 2.22	90.56 ± 0.53	90.29 ± 1.44	85.81 ± 1.65	93.23 ± 1.13	93.71 ± 0.75	94.43 ± 1.05	95.22 ± 1.33	$\mathbf{95.85 \pm 0.71}$
12	69.80 ± 8.02	81.46 ± 4.46	87.31 ± 3.35	88.57 ± 5.12	91.36 ± 3.29	89.83 ± 2.07	90.82 ± 3.13	90.15 ± 2.67	$\mathbf{92.30 \pm 1.98}$
13	97.70 ± 1.11	98.91 ± 0.76	98.51 ± 1.19	98.43 ± 1.23	98.48 ± 1.34	97.83 ± 1.44	$\mathbf{99.46 \pm 1.79}$	95.32 ± 0.35	98.59 ± 1.02
14	98.04 ± 0.42	98.98 ± 0.68	98.59 ± 0.82	95.01 ± 1.39	99.24 ± 0.46	99.51 ± 0.39	99.12 ± 0.45	96.26 ± 3.26	$\mathbf{99.90 \pm 0.12}$
15	89.87 ± 1.22	75.80 ± 5.25	81.46 ± 5.16	88.79 ± 3.92	94.91 ± 2.50	$\mathbf{96.72 \pm 2.41}$	96.55 ± 1.37	93.11 ± 1.05	95.86 ± 4.10
16	96.38 ± 4.28	$\mathbf{97.94 \pm 2.32}$	81.02 ± 6.09	96.62 ± 1.68	95.78 ± 3.90	96.62 ± 2.59	96.39 ± 3.74	82.90 ± 2.29	96.02 ± 3.45
OA	87.07 ± 1.81	86.11 ± 2.63	88.58 ± 2.65	87.55 ± 2.96	94.54 ± 1.64	95.58 ± 1.34	95.87 ± 0.50	95.74 ± 0.69	$\mathbf{96.69 \pm 1.25}$
AA	88.04 ± 2.78	85.42 ± 4.98	88.11 ± 3.55	85.44 ± 3.41	95.13 ± 2.49	96.08 ± 1.85	96.16 ± 0.57	88.13 ± 1.01	$\mathbf{97.14 \pm 1.48}$
κ	85.18 ± 2.94	84.01 ± 3.74	86.90 ± 3.76	85.80 ± 3.10	93.77 ± 3.73	94.96 ± 2.38	95.12 ± 1.57	95.26 ± 2.31	$\mathbf{96.22 \pm 2.28}$

of α on the Indian Pines dataset. The experimental results indicate that the optimal parameter configuration for Indian Pines is $(\alpha, \beta, \gamma) = (0.0001, 0.0001, 0.01)$. For the Salinas, HeiHe, and WHU-Hi-HanChuan datasets are $(0.0001, 0.001, 0.01)$, $(0.1, 0.1, 0.01)$, and $(1, 0.01, 0.1)$, respectively. Overall, for relatively regular agricultural datasets, smaller graph-related parameters (α and β) are sufficient to achieve high performance, indicating that spectral discrimination predominantly drives classification. In contrast, for more structurally complex mixed-scene datasets, larger α and β are generally required to effectively capture intrinsic geometrical structures. Meanwhile, the parameter γ is relatively large across all four datasets, enhancing the model's robustness against noise. Fig. 3 shows the variation of OA on four HSI datasets with different dimensions. Fig. 4 shows the variation of OA on four HSI datasets with respect to parameter k. Based on these results, the parameter $\tilde{d}$, which represents the feature dimensionality, was set to 30. The parameter k was set to 3, while m is the number of anchors is $2^k - 1$. All parameters used in the experiments were chosen with reference to relevant literature. Before the experiments, the datasets were processed using a mean filter.

C. Comparison of Classification Performance

1) Comparison With ML Methods: Tables IV–VII present a quantitative comparison of classification metrics obtained

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TABLE V
CLASSIFICATION ACCURACY (%) OF THE SALINAS DATASET WITH DIFFERENT CLASSIFICATION METHODS

Class	RLR	DLRP	DRR	OAR	SSLA-RLR	PSAE-DMLSR	TDRLR	LatLRE	**FAGRLR**
1	99.23 ± 1.25	99.14 ± 0.61	99.66 ± 0.87	$\mathbf{99.89 \pm 0.12}$	98.01 ± 1.85	99.46 ± 1.05	99.28 ± 0.60	99.58 ± 0.74	99.73 ± 0.50
2	99.98 ± 0.03	99.59 ± 0.15	99.91 ± 0.13	96.84 ± 0.97	99.63 ± 0.41	$\mathbf{99.98 \pm 0.35}$	99.89 ± 0.13	99.70 ± 1.09	99.97 ± 0.14
3	95.86 ± 2.53	98.74 ± 0.60	99.02 ± 0.67	83.29 ± 1.14	98.20 ± 1.32	99.54 ± 0.14	$\mathbf{99.94 \pm 0.59}$	98.57 ± 1.17	99.80 ± 0.28
4	97.88 ± 0.72	98.81 ± 0.81	95.61 ± 2.27	95.56 ± 1.97	96.89 ± 1.83	96.14 ± 2.40	$\mathbf{96.99 \pm 3.28}$	95.98 ± 1.24	96.42 ± 1.17
5	96.36 ± 0.81	97.73 ± 1.05	93.03 ± 3.06	98.59 ± 0.55	97.27 ± 1.29	98.80 ± 0.89	96.15 ± 0.21	$\mathbf{99.89 \pm 1.32}$	98.54 ± 0.87
6	99.88 ± 0.22	99.48 ± 0.27	99.90 ± 0.05	$\mathbf{99.98 \pm 0.03}$	99.74 ± 0.73	99.81 ± 0.10	99.55 ± 0.74	98.35 ± 0.26	99.75 ± 0.35
7	99.42 ± 0.16	99.61 ± 0.36	97.82 ± 1.33	$\mathbf{99.79 \pm 0.14}$	99.63 ± 0.23	99.70 ± 0.02	98.97 ± 1.36	99.08 ± 1.21	99.42 ± 0.37
8	93.25 ± 1.10	89.12 ± 1.13	87.94 ± 1.66	92.25 ± 0.82	85.67 ± 2.92	89.94 ± 2.17	93.06 ± 0.81	91.46 ± 2.11	$\mathbf{93.85 \pm 1.24}$
9	99.99 ± 0.02	99.18 ± 0.26	99.71 ± 0.12	99.77 ± 0.19	99.33 ± 0.73	99.75 ± 0.44	$\mathbf{99.99 \pm 2.03}$	95.18 ± 1.14	99.56 ± 0.70
10	89.73 ± 3.54	90.29 ± 2.83	93.18 ± 2.43	92.68 ± 0.91	94.20 ± 2.08	95.73 ± 1.65	96.00 ± 0.87	93.87 ± 2.25	$\mathbf{96.97 \pm 1.40}$
11	97.61 ± 1.53	90.79 ± 1.88	$\mathbf{99.82 \pm 0.26}$	98.85 ± 0.59	98.78 ± 0.93	98.61 ± 0.58	99.74 ± 0.48	97.85 ± 0.55	99.30 ± 0.70
12	98.98 ± 0.88	99.49 ± 0.46	94.43 ± 3.39	81.42 ± 4.45	95.73 ± 2.66	$\mathbf{99.50 \pm 1.74}$	98.62 ± 3.47	99.14 ± 0.98	98.90 ± 1.07
13	99.22 ± 1.65	97.90 ± 0.75	97.01 ± 3.15	84.58 ± 3.56	93.23 ± 2.83	94.85 ± 2.46	98.95 ± 3.22	$\mathbf{99.26 \pm 3.25}$	96.84 ± 1.90
14	94.76 ± 2.13	92.77 ± 2.98	87.40 ± 2.34	90.07 ± 3.53	95.15 ± 1.98	95.21 ± 1.67	95.11 ± 1.58	96.71 ± 0.32	$\mathbf{97.69 \pm 1.26}$
15	54.89 ± 2.27	55.41 ± 4.04	72.07 ± 4.32	76.97 ± 3.57	82.16 ± 2.07	82.79 ± 1.50	81.20 ± 2.12	83.74 ± 2.21	$\mathbf{87.67 \pm 1.40}$
16	95.61 ± 2.72	97.69 ± 1.53	98.13 ± 2.24	97.79 ± 1.96	97.60 ± 1.83	99.22 ± 1.46	99.88 ± 1.72	$\mathbf{99.90 \pm 0.32}$	98.56 ± 1.03
OA	91.10 ± 0.96	90.30 ± 1.06	92.10 ± 0.43	92.58 ± 0.63	93.30 ± 1.51	94.82 ± 1.02	95.24 ± 2.23	95.71 ± 0.82	$\mathbf{96.39 \pm 0.85}$
AA	94.54 ± 1.64	94.11 ± 1.23	94.66 ± 1.76	93.02 ± 1.53	95.70 ± 1.70	96.81 ± 1.16	97.08 ± 0.47	96.76 ± 2.35	$\mathbf{97.69 \pm 1.31}$
κ	90.05 ± 2.60	89.16 ± 1.78	91.20 ± 2.98	91.73 ± 2.07	92.54 ± 2.64	94.23 ± 2.03	94.69 ± 0.18	95.28 ± 3.19	$\mathbf{95.98 \pm 1.89}$

TABLE VI
CLASSIFICATION ACCURACY (%) OF THE HEIHE DATASET WITH DIFFERENT CLASSIFICATION METHODS

Class	RLR	DLRP	DRR	OAR	SSLA-RLR	PSAE-DMLSR	TDRLR	LatLRE	**FAGRLR**
1	89.27 ± 3.04	89.66 ± 3.14	95.72 ± 0.47	98.65 ± 0.32	98.61 ± 0.16	98.79 ± 0.27	98.23 ± 1.46	96.41 ± 0.77	$\mathbf{99.58 \pm 0.59}$
2	94.43 ± 2.73	96.60 ± 1.08	96.96 ± 0.62	98.43 ± 0.10	98.61 ± 0.21	98.71 ± 0.25	98.42 ± 0.99	98.42 ± 1.16	$\mathbf{99.43 \pm 0.15}$
3	89.01 ± 2.28	87.27 ± 5.40	92.93 ± 1.35	96.88 ± 0.36	97.46 ± 0.27	98.06 ± 0.48	99.55 ± 0.78	99.02 ± 0.74	$\mathbf{99.87 \pm 0.17}$
4	85.10 ± 2.42	91.51 ± 3.01	83.73 ± 1.33	89.43 ± 0.85	90.95 ± 1.12	93.67 ± 1.41	93.36 ± 0.42	95.14 ± 0.21	$\mathbf{96.03 \pm 0.91}$
5	95.12 ± 1.25	94.17 ± 1.34	93.77 ± 2.58	96.78 ± 0.53	97.03 ± 0.53	95.15 ± 1.28	95.98 ± 0.66	96.77 ± 2.02	$\mathbf{99.00 \pm 0.61}$
6	64.73 ± 4.07	59.09 ± 0.40	65.78 ± 2.72	59.21 ± 3.72	54.39 ± 4.17	63.01 ± 2.43	$\mathbf{75.43 \pm 1.39}$	69.65 ± 0.35	73.66 ± 3.90
7	33.91 ± 5.66	40.77 ± 2.65	43.52 ± 4.77	67.21 ± 2.81	64.14 ± 5.67	72.53 ± 4.75	74.12 ± 3.91	75.16 ± 2.16	$\mathbf{76.51 \pm 6.50}$
8	97.60 ± 1.29	96.04 ± 2.31	84.31 ± 3.59	95.81 ± 0.37	95.17 ± 1.31	94.37 ± 2.06	97.22 ± 1.54	$\mathbf{99.12 \pm 0.64}$	97.68 ± 0.86
OA	90.72 ± 1.55	91.52 ± 1.73	93.51 ± 1.28	96.56 ± 0.85	96.72 ± 0.92	97.27 ± 1.13	96.46 ± 0.15	96.72 ± 0.97	$\mathbf{98.83 \pm 1.21}$
AA	81.14 ± 2.84	81.89 ± 1.82	82.09 ± 2.08	87.80 ± 1.13	87.04 ± 1.87	89.28 ± 1.61	91.53 ± 0.42	91.21 ± 0.61	$\mathbf{92.72 \pm 1.75}$
κ	87.56 ± 3.51	88.61 ± 2.27	91.05 ± 3.38	95.26 ± 2.21	95.48 ± 2.17	96.24 ± 2.39	95.13 ± 0.28	95.28 ± 1.52	$\mathbf{98.39 \pm 1.92}$

by various methods, including EA, OA, AA, and κ, for the Indian Pines, Salinas, HeiHe, and WHU-Hi-HanChuan datasets, respectively. In addition, Figs. 5–8 provide visual comparisons of the corresponding classification maps.

In general, the proposed FAGRLR demonstrates outstanding classification performance across all datasets. Table IV presents the classification results for the Indian Pines dataset. Conventional methods, such as DLRP, exhibit relatively poor performance, which can be attributed to their limited ability to capture the global structure and complex nonlinear characteristics inherent in high-dimensional data. When compared to other LR-based methods, including RLR, DRR, OAR, SSLA-RLR, PSAE-DMLSR, and TDRLR, FAGRLR achieves a substantial improvement in classification accuracy. Specifically, the OA obtained by FAGRLR exceeds those of RLR, DRR, OAR, SSLA-RLR, PSAE-DMLSR, TDRLR, and LatLRE by 9.62%, 8.11%, 9.14%, 2.15%, 1.11%, 0.82%, and 0.95%, respectively. It demonstrates the effectiveness of FAGRLR in handling nonlinear data and preserving the local structure of high-dimensional feature spaces. Moreover, the improvements in both AA and κ further confirm the effectiveness of the proposed method for HSI classification. Fig. 5 shows the classification maps for all evaluated methods on the Indian Pines dataset.

As shown in Table V, the proposed method achieves the best performance in terms of the OA, AA, and κ on Salinas dataset. Compared to RLR, DLRP, DRR, OAR, SSLA-RLR, PSAE-DMLSR, TDRLR, and LatLRE, FAGRLR achieves an improvement of at least 0.68%, highlighting the importance of local geometric structure preservation in high-dimensional data. Fig. 6 illustrates that the land-cover information obtained by FAGRLR is closest to the original ground truth data. For the HeiHe and WHU-Hi-HanChuan datasets, the classification results presented in Tables VI and VII confirm the superior performance of the proposed method. On the HeiHe dataset, it performs particularly well in classifying certain categories, such as class 6 (chicory seedlings) and class 7 (watermelon). On the WHU-Hi-HanChuan dataset, FAGRLR consistently outperforms other methods, achieving improvements of 1.27%, 2.1%, and 0.87% in terms of OA, AA, and the κ coefficient, respectively. The detailed results are shown in Figs. 7 and 8.

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TABLE VII
CLASSIFICATION ACCURACY (%) OF THE WHU-HI-HANCHUAN DATASET WITH DIFFERENT CLASSIFICATION METHODS

Class	RLR	DLRP	DRR	OAR	SSLA-RLR	PSAE-DMLSR	TDRLR	LatLRE	FAGRLR
1	69.39 ± 9.01	82.41 ± 7.16	87.85 ± 4.02	86.56 ± 2.64	86.31 ± 5.41	87.82 ± 2.26	$\mathbf{91.21 \pm 1.07}$	89.45 ± 3.28	90.15 ± 2.84
2	74.45 ± 5.88	74.02 ± 2.82	75.41 ± 2.84	77.23 ± 4.12	74.58 ± 2.83	84.99 ± 2.60	89.82 ± 0.97	$\mathbf{90.16 \pm 2.12}$	88.92 ± 1.59
3	51.46 ± 3.72	84.26 ± 6.68	86.94 ± 6.25	79.27 ± 5.14	76.60 ± 4.59	89.14 ± 6.89	90.03 ± 0.31	93.74 ± 1.55	$\mathbf{96.75 \pm 3.60}$
4	99.77 ± 0.35	98.13 ± 0.49	97.47 ± 0.56	97.67 ± 0.63	98.54 ± 1.17	98.64 ± 0.96	98.75 ± 0.12	98.65 ± 1.15	$\mathbf{99.80 \pm 0.71}$
5	97.65 ± 0.93	98.09 ± 2.22	97.28 ± 4.56	97.06 ± 1.24	98.81 ± 1.61	99.90 ± 0.25	87.96 ± 2.01	95.72 ± 1.16	$\mathbf{99.95 \pm 1.08}$
6	70.74 ± 8.92	53.69 ± 7.13	56.12 ± 8.16	61.06 ± 7.38	70.22 ± 6.48	66.38 ± 8.72	83.71 ± 3.58	85.26 ± 0.88	$\mathbf{87.92 \pm 2.95}$
7	$\mathbf{97.09 \pm 7.21}$	92.87 ± 2.04	93.93 ± 2.13	90.77 ± 1.72	94.51 ± 1.40	95.51 ± 3.67	86.72 ± 2.31	90.57 ± 0.32	96.56 ± 3.29
8	45.98 ± 5.12	51.32 ± 10.13	51.96 ± 8.12	65.49 ± 4.90	70.14 ± 3.35	69.89 ± 6.38	$\mathbf{84.14 \pm 1.18}$	82.72 ± 0.58	81.53 ± 4.59
9	64.39 ± 6.77	61.46 ± 8.99	61.88 ± 9.72	75.81 ± 6.16	74.33 ± 4.71	85.03 ± 5.37	86.53 ± 1.12	79.01 ± 1.75	$\mathbf{89.37 \pm 2.25}$
10	98.20 ± 0.14	95.58 ± 0.94	95.01 ± 2.04	95.74 ± 0.84	97.64 ± 0.70	$\mathbf{98.89 \pm 0.99}$	95.21 ± 0.27	96.45 ± 0.65	98.43 ± 0.48
11	63.35 ± 5.37	93.02 ± 2.61	94.33 ± 2.10	90.61 ± 3.19	$\mathbf{95.44 \pm 3.24}$	89.85 ± 4.22	91.75 ± 0.82	94.88 ± 2.26	95.12 ± 2.57
12	69.37 ± 6.32	65.61 ± 4.90	67.44 ± 2.74	70.21 ± 4.25	76.82 ± 6.01	75.34 ± 5.25	86.89 ± 4.75	93.09 ± 0.82	$\mathbf{97.76 \pm 3.68}$
13	35.75 ± 2.74	67.27 ± 4.18	67.99 ± 4.93	61.34 ± 7.70	65.35 ± 2.39	74.25 ± 4.30	$\mathbf{76.39 \pm 4.05}$	75.42 ± 2.45	73.52 ± 2.26
14	31.29 ± 3.50	57.13 ± 4.82	61.71 ± 7.33	67.25 ± 5.88	72.79 ± 6.98	66.07 ± 3.76	$\mathbf{91.90 \pm 2.67}$	80.35 ± 0.73	83.69 ± 1.77
15	85.38 ± 2.12	89.66 ± 2.54	89.64 ± 3.91	89.08 ± 4.10	86.20 ± 6.76	90.68 ± 3.63	86.79 ± 4.63	$\mathbf{93.71 \pm 0.22}$	93.08 ± 1.72
16	95.88 ± 3.45	94.67 ± 3.91	94.23 ± 3.67	95.72 ± 2.21	94.77 ± 2.53	$\mathbf{98.60 \pm 5.64}$	93.85 ± 0.35	90.89 ± 0.50	90.98 ± 2.03
OA	72.00 ± 2.72	81.20 ± 3.88	82.80 ± 3.39	84.32 ± 2.84	85.22 ± 2.75	87.88 ± 3.32	88.15 ± 0.31	88.72 ± 0.85	$\mathbf{89.99 \pm 3.38}$
AA	71.88 ± 3.74	78.70 ± 4.07	79.95 ± 4.56	81.30 ± 3.88	83.32 ± 4.59	85.68 ± 4.06	88.85 ± 0.78	89.37 ± 0.44	$\mathbf{91.47 \pm 3.22}$
κ	67.86 ± 2.98	78.25 ± 4.11	80.08 ± 3.15	81.81 ± 4.95	82.88 ± 4.99	85.90 ± 3.71	86.43 ± 0.32	87.55 ± 1.25	$\mathbf{88.42 \pm 4.14}$

TABLE VIII
COMPARISON OF CLASSIFICATION ACCURACY (%) AND COMPUTATIONAL COST ON FOUR HSI DATASETS

	Metrics	WFCG	GPF-NET	GraphGST	3DSS-Mamba	DiffFormer	MambaHSI+	FAGRLR
Indian Pines	OA	93.51 ± 1.64	95.14 ± 1.99	95.47 ± 0.16	94.91 ± 3.42	93.96 ± 2.16	$\mathbf{96.87 \pm 0.46}$	96.69 ± 1.25
	AA	90.11 ± 3.58	93.85 ± 0.98	94.35 ± 0.61	89.70 ± 2.07	91.28 ± 3.47	96.68 ± 2.16	$\mathbf{97.14 \pm 1.48}$
	κ	88.43 ± 2.16	92.69 ± 2.08	94.96 ± 1.03	94.25 ± 2.11	88.74 ± 4.71	95.99 ± 0.62	$\mathbf{96.22 \pm 2.28}$
	Tr(s)	15.536	23.714	59.126	3.479	78.64	35.61	**0.257**
	Te(s)	0.119	0.194	3.548	0.641	1.51	1.26	**0.001**
Salinas	OA	92.68 ± 1.31	94.51 ± 0.90	95.74 ± 0.69	93.16 ± 1.47	92.51 ± 1.77	96.12 ± 1.12	$\mathbf{96.39 \pm 0.85}$
	AA	93.55 ± 1.94	95.16 ± 0.94	96.89 ± 0.99	96.31 ± 0.95	94.65 ± 2.25	97.21 ± 2.71	$\mathbf{97.69 \pm 1.31}$
	κ	91.72 ± 0.61	93.15 ± 0.86	94.48 ± 0.82	92.40 ± 2.61	90.68 ± 1.15	$\mathbf{96.06 \pm 1.72}$	95.98 ± 1.89
	Tr(s)	43.628	65.493	71.998	1.631	69.74	25.82	**0.145**
	Te(s)	0.822	0.414	9.624	0.257	1.42	1.55	**0.007**
HeiHe	OA	91.47 ± 0.59	92.48 ± 1.63	95.02 ± 0.35	95.76 ± 0.81	93.24 ± 2.75	96.72 ± 3.18	$\mathbf{98.83 \pm 1.21}$
	AA	89.74 ± 0.77	88.16 ± 2.69	90.36 ± 0.52	93.58 ± 2.37	88.76 ± 1.26	92.05 ± 2.08	$\mathbf{92.72 \pm 1.75}$
	κ	90.66 ± 1.88	91.85 ± 1.12	91.69 ± 1.28	94.16 ± 0.66	90.45 ± 0.99	93.88 ± 0.56	$\mathbf{98.39 \pm 1.92}$
	Tr(s)	13.526	10.625	50.948	1.957	67.35	28.65	**0.271**
	Te(s)	0.391	0.463	12.135	0.166	5.69	2.98	**0.006**
WHU-Hi-HanChuan	OA	86.19 ± 2.12	86.94 ± 1.63	87.45 ± 1.84	87.88 ± 4.34	85.29 ± 1.71	88.76 ± 4.45	$\mathbf{89.99 \pm 3.38}$
	AA	87.64 ± 1.94	88.41 ± 0.82	89.61 ± 0.80	88.07 ± 5.17	86.36 ± 0.87	91.08 ± 1.74	$\mathbf{91.47 \pm 3.22}$
	κ	86.61 ± 1.34	87.55 ± 1.17	89.15 ± 0.39	86.05 ± 5.06	84.91 ± 2.25	$\mathbf{88.64 \pm 2.16}$	88.42 ± 4.14
	Tr(s)	150.325	107.116	202.663	8.273	226.61	84.15	**0.114**
	Te(s)	0.628	1.035	18.253	1.716	65.16	2.38	**0.023**

2) Comparison With DL Methods: To further assess the effectiveness of the proposed method, we also compare FAGRLR with several representative DL approaches, including WFCG [23], GPF-NET [24], GraphGST [27], 3DSS-Mamba [32], DiffFormer [29], and MambaHSI+ [33]. As shown in Table VIII, although DL methods generally achieve strong classification performance in some scenarios, they still exhibit certain limitations in terms of computational complexity. In contrast, the proposed FAGRLR consistently delivers superior performance across all datasets while requiring substantially lower computational resources.

D. Ablation Experiment

In this section, an ablation experiment is conducted to analyze the impact of each component on the model's classification performance. A series of comparative experiments is performed based on the four datasets used in this study. The FAGRLR version without AG, label relaxation (LAR), and $L_{2,1}$-norm constraint ($L_{2,1}$-NC) components serve as the baseline models for comparison. The classification performance is evaluated using the average OA, AA, and κ indices across multiple datasets under different conditions. Experimental results, summarized in Table IX, indicate that incorporating these three components significantly improves the classification accuracy of HSI classification algorithms. Specifically, on the Indian Pines dataset, introducing only the AG component leads to improvements of approximately 7%, 9%, and 7% in OA, AA, and κ, respectively. When only the LAR component is incorporated, the corresponding metrics improve by about 7%, 12%, and 9%. In the case of

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TABLE IX
ABLATION ANALYSIS FOR THE PROPOSED METHOD

Baseline	AG	LAR	$L_{2,1}$-NC	Indian Pines			Salinas		
				OA	AA	κ	OA	AA	κ
✓	✗	✗	✗	79.17 ± 0.56	75.22 ± 0.24	76.08 ± 0.63	83.64 ± 0.98	85.41 ± 0.95	81.74 ± 1.09
✓	✓	✗	✗	85.76 ± 0.14	84.49 ± 0.51	83.69 ± 0.16	90.04 ± 0.62	93.51 ± 0.14	88.89 ± 0.70
✓	✗	✓	✗	87.07 ± 1.81	88.04 ± 2.78	85.18 ± 2.94	91.10 ± 0.96	94.54 ± 1.64	90.05 ± 2.60
✓	✗	✗	✓	86.70 ± 0.36	86.54 ± 1.25	84.76 ± 0.39	91.00 ± 0.22	92.82 ± 1.15	89.96 ± 0.25
✓	✓	✓	✗	93.88 ± 0.33	94.86 ± 0.42	93.02 ± 0.38	94.38 ± 0.09	96.91 ± 0.22	93.75 ± 0.13
✓	✓	✗	✓	90.44 ± 0.39	90.81 ± 0.94	89.07 ± 0.46	93.61 ± 0.21	96.03 ± 0.16	92.88 ± 0.24
✓	✗	✓	✓	88.45 ± 0.37	90.21 ± 1.03	86.85 ± 0.43	92.06 ± 0.08	95.11 ± 0.27	91.15 ± 0.09
✓	✓	✓	✓	$\mathbf{96.69 \pm 1.25}$	$\mathbf{97.14 \pm 1.48}$	$\mathbf{96.22 \pm 2.28}$	$\mathbf{96.39 \pm 0.85}$	$\mathbf{97.69 \pm 1.31}$	$\mathbf{95.98 \pm 1.89}$
Baseline	AG	LAR	$L_{2,1}$-NC	HeiHe			WHU-Hi-HanChuan		
				OA	AA	κ	OA	AA	κ
✓	✗	✗	✗	86.31 ± 2.60	81.65 ± 3.44	81.65 ± 3.45	70.96 ± 1.63	72.73 ± 1.28	67.02 ± 1.78
✓	✓	✗	✗	95.10 ± 0.51	86.71 ± 1.70	94.76 ± 0.44	83.89 ± 0.76	85.87 ± 0.45	81.47 ± 0.84
✓	✗	✓	✗	90.72 ± 1.55	81.14 ± 2.84	87.56 ± 3.51	72.00 ± 2.72	70.63 ± 3.74	67.86 ± 2.98
✓	✗	✗	✓	96.51 ± 0.18	88.83 ± 1.52	95.19 ± 0.21	80.85 ± 1.53	83.68 ± 0.88	78.09 ± 1.71
✓	✓	✓	✗	96.78 ± 0.11	84.03 ± 1.24	95.59 ± 0.15	87.35 ± 0.45	91.18 ± 0.12	85.44 ± 0.51
✓	✓	✗	✓	95.03 ± 0.14	83.95 ± 1.09	93.15 ± 0.20	85.54 ± 0.95	87.54 ± 0.35	83.64 ± 0.31
✓	✗	✓	✓	96.66 ± 0.28	82.74 ± 1.21	95.40 ± 0.39	84.31 ± 0.96	89.29 ± 0.28	82.06 ± 1.06
✓	✓	✓	✓	$\mathbf{98.83 \pm 1.21}$	$\mathbf{92.72 \pm 1.75}$	$\mathbf{98.39 \pm 1.92}$	$\mathbf{89.99 \pm 3.38}$	$\mathbf{91.47 \pm 3.22}$	$\mathbf{88.42 \pm 4.14}$

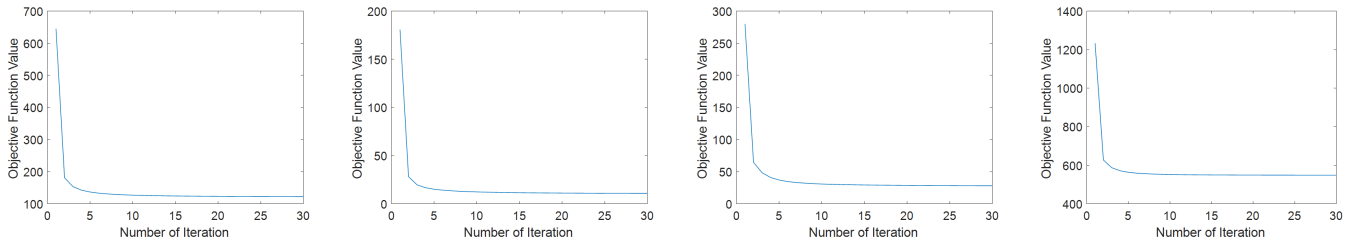


Fig. 9. Convergence curvesof FAGRLR on the four datasets. (a) Indian Pines. (b) Salinas. (c) HeiHe. (d) WHU-Hi-HanChuan.

introducing only the $L_{2,1}$-NC component, the three indicators increase by approximately 7%, 11%, and 8%. To further assess the necessity of each component, we additionally constructed three variants in which one component was removed, while the other two were retained. Experimental results show that removing any single module leads to noticeable performance degradation. Specifically, on the Indian Pines dataset, eliminating AG results in drops of approximately 8%, 7%, and 10% in OA, AA, and κ, respectively, while removing LAR or $L_{2,1}$-NC also causes a certain degree of performance decline. Similar conclusions can also be drawn from the experimental results on the Salinas, HeiHe, and WHU-Hi-HanChuan datasets, as shown in Table IX. Therefore, it can be concluded that the AG, LAR, and $L_{2,1}$-NC components play a crucial role in the FAGRLR, significantly enhancing the robustness of the classification algorithm.

E. Convergence Experiment

In the earlier discussion, a theoretical analysis of the convergence behavior of the proposed algorithm is presented. To empirically validate the effectiveness of the proposed method, convergence experiments were conducted across four HSI datasets, adhering to the previously outlined experimental configurations. The convergence curves for the proposed method are presented in Fig. 9, where it is evident that as the number of iterations increases, the objective function value progressively decreases and eventually stabilizes, indicating convergence. The experimental results corroborate the theoretical analysis, demonstrating the effectiveness of the proposed algorithm. Notably, the convergence curve reaches stability after approximately 15 iterations, indicating that the FAGRLR converges rapidly and thus underscores its practical applicability.

IV. CONCLUSION

In this article, a novel method named FAGRLR is proposed for HSI classification, which integrates a fast AG into an LR model to effectively capture the local structural characteristics of HSIs. A label relaxation strategy is employed to enhance the discriminative capability. In addition, an $L_{2,1}$-norm regularization is imposed on the transformation matrix

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to enhance robustness against noise and outliers. To evaluate the effectiveness of the proposed method, comprehensive experiments were conducted on four widely used hyperspectral datasets. The results demonstrate that FAGRLR consistently achieves superior classification performance, exhibiting both strong robustness and enhanced discriminative capability compared with existing methods. Nevertheless, since FAGRLR is essentially built upon an LR framework, its ability to model highly nonlinear spectral variations remains limited, which will be further investigated in future work.

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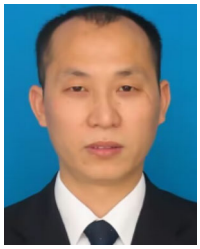
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